

AERO50001 Aerodynamics 2 Problem Set #3

Tutorial 1

3.1 Rounded leading-edge aerofoil

☆☆☆ 10-15 mins

A symmetrical aerofoil with a rounded leading edge is flying at $M = 3$ at zero incidence and at sea level conditions ($T_1 = 288 \text{ K}$, $p_1 = 1.013 \cdot 10^5 \text{ N/m}^2$).

Hint: a curved shock forms in front of the aerofoil, however just in front of the nose it is a normal shock.

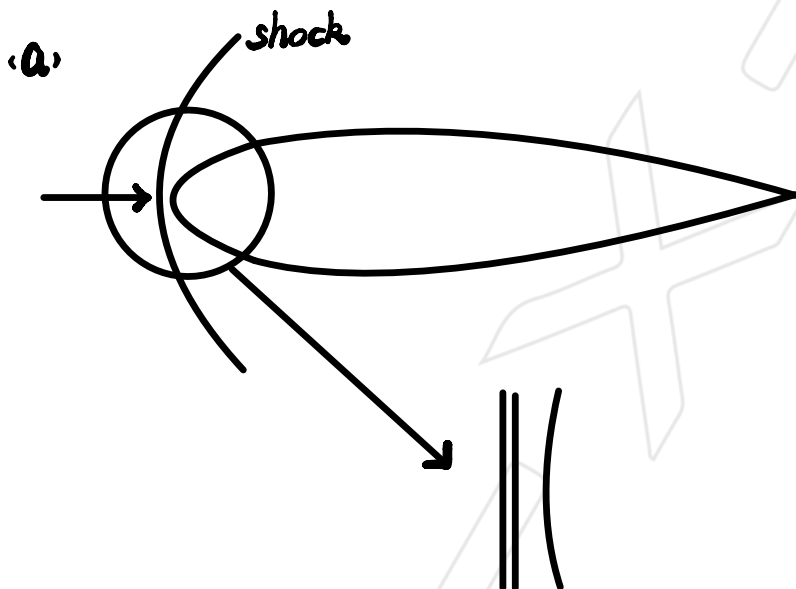
(a) Calculate the temperature and pressure at the nose of the aerofoil.

Hint: the nose of the aerofoil is a stagnation point.

Note: to enter numbers in scientific format, use format 1.5e6, of instance]

(b) Determine the pressure coefficient at the nose of the aerofoil.

(c) What is the pressure coefficient at a point on the aerofoil where the local Mach number is equal to 1?



$$M = 3$$

According to the table

$$\frac{p_2}{p_1} = 10.33, \quad \frac{\rho_2}{\rho_1} = 3.857$$

$$M_2 = 0.4752$$

$$\therefore p_2 = 10.33 p_1 = 1.046 \times 10^6 \text{ Pa}$$

$$\therefore \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right) \left(\frac{\rho_1}{\rho_2}\right)$$

$$\therefore T_2 = \frac{10.33}{3.857} T_1 = 771.3 \text{ K}$$

$$\therefore T_{02} = T_2 \left(1 + \frac{\gamma-1}{2} M^2\right) = 806.17 \text{ K}$$

$$\therefore p_{02} = p_2 \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}} = 1221389 \text{ Pa} = 1.22 \times 10^6 \text{ Pa}$$

$$b) \quad C_p = \frac{p - p_{\infty}}{\frac{1}{2} \rho U_{\infty}^2} = \frac{p - p_{\infty}}{\frac{1}{2} \frac{p_{\infty}}{RT_{\infty}} (M_{\infty} \sqrt{\gamma RT_{\infty}})^2} = \left(\frac{p}{p_{\infty}} - 1\right) \frac{1}{\frac{1}{2} \gamma (M_{\infty})^2}$$

$$\therefore C_p \text{ at nose} = \left(\frac{p_{02}}{p_1} - 1 \right) \frac{1}{\frac{1}{2} \rho M_1^2} = 1.755$$

$$\text{c. } C_p \text{ at aerofoil} = \left(\frac{p_2}{p_1} - 1 \right) \frac{1}{\frac{1}{2} \rho M_1^2}$$

$$= \left(\frac{p_2}{p_{02}} \frac{p_{02}}{p_1} - 1 \right) \frac{1}{\frac{1}{2} \rho M_1^2}$$

$$= 0.8523$$

3.2 Maximum mass flow rate

★★☆ 10-20 mins

Find an expression for the **maximum** mass flow rate $\dot{m}$ through a convergent-divergent nozzle as a function of γ , the throat area A_t , and the reservoir density and speed of sound, ρ_0 and a_0 respectively.

[Note: for exponents, use ** instead of ^.]

$$\dot{m}^* = \rho^* u^* A_t = \rho^* M a^* A_t \stackrel{M=1}{=} \rho^* a^* A_t$$

$$= \rho_0 \left(1 + \frac{\gamma-1}{2}\right)^{\frac{1}{\gamma-1}} a^* A_t = \rho_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} a^* A_t$$

$$a^* = \sqrt{\gamma R T^*} = \sqrt{\gamma R T_0 \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1}} \stackrel{M=1}{=} a_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{2}}$$

$$\therefore \dot{m} = \rho_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} a_0 \left(\frac{2}{\gamma+1}\right)^{\frac{1}{2}} A_t$$

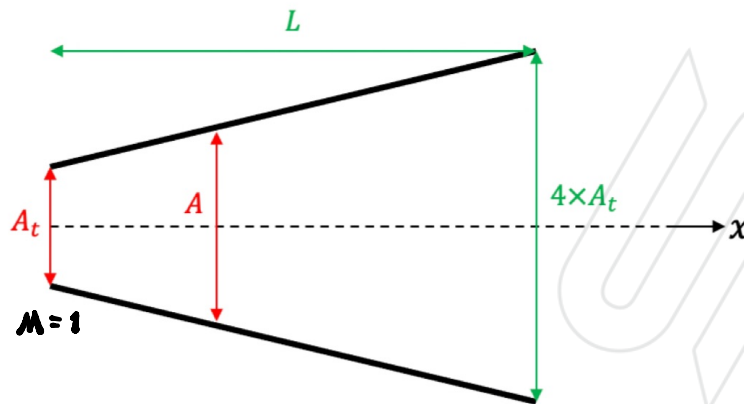
$$= \rho_0 a_0 A_t \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

3.3 Conical supersonic nozzles

★★☆ 15-25 mins

Consider a conical, supersonic nozzle, in which the area expands from A_t at the sonic throat to $4A_t$ at the exit, which is a distance L from the throat.

Assume air with $\gamma = 1.4$.



- Find an expression for the diameter of any section along the nozzle D as a function of the diameter at the throat D_t and the non-dimensional distance from the throat x/L .
- Find an expression for the area of any section along the nozzle A as a function of the area at the throat A_t and the non-dimensional distance from the throat x/L .
- Using the expressions you have found, calculate the area and Mach number at a section located a distance $0.6L$ from the throat.
- OPTIONAL: plot (manually or using MATLAB) the variation of area and Mach number along the nozzle. You may assume any radius at the throat, i.e. 1.

$$(a) A_t = \frac{\pi D_t^2}{4}$$

$$4 \times A_t = \frac{\pi D^2}{4} = \pi D_t^2$$

$$\therefore \text{At the } x=L, D^2 = 4D_t^2 \quad \therefore D = 2D_t$$

$$\therefore D = \alpha \frac{x}{L} + \beta$$

$$\text{At } \frac{x}{L} = 0, D = D_t \quad \therefore \beta = D_t$$

$$\text{At } \frac{x}{L} = 1, D = 2D_t \quad \therefore \alpha = D_t$$

$$\therefore D = D_t \frac{x}{L} + D_t = D_t \left(\frac{x}{L} + 1 \right)$$

$$(b) A = \frac{\pi D^2}{4} = \frac{\pi [D_t (\frac{x}{L} + 1)]^2}{4} = \frac{\pi D_t^2}{4} (\frac{x}{L} + 1)^2 = A_t (\frac{x}{L} + 1)^2$$

$$(c) \text{ At } \frac{x}{L} = 0.6, A = 2.56 A_t$$

$$\therefore \frac{A}{A_t} = 2.56$$

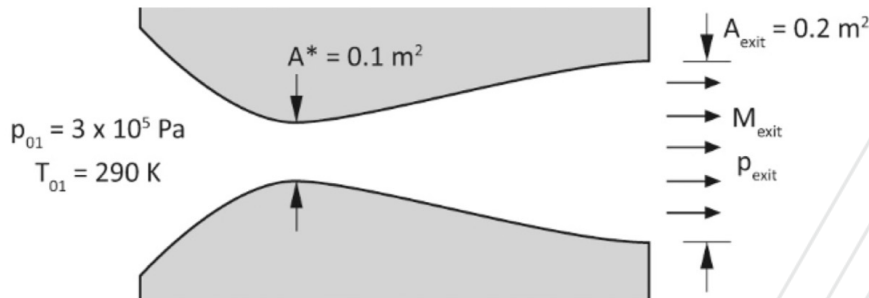
$$\therefore M = 2.45$$

(e) ...

3.4 Convergent-divergent nozzles

★★★ 20-30 mins

The figure below shows a convergent-divergent nozzle. It is observed that the flow (air, $\gamma = 1.4$) exits the nozzle as a perfectly expanded (isentropic) supersonic flow.



- Determine the mass flow rate through the nozzle.
- Determine the Mach number at the nozzle exit (M_{exit}).
- Determine the pressure at the nozzle exit (p_{exit}).
- The pressure at the nozzle exit is gradually increased from its original value. For what range of values of p_{exit} would the mass flow rate through the nozzle remain unchanged?

$$\begin{aligned}
 \text{(a)} \quad \dot{m}^* &= \rho^* u^* A_t \stackrel{M=1}{=} \frac{p^*}{RT^*} \sqrt{\gamma RT^*} A_t = \frac{p^*}{\sqrt{RT^*}} \sqrt{\gamma} A_t \\
 &= p_0 \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} \frac{1}{\sqrt{RT_0}} \left(\frac{2}{\gamma+1} \right)^{-\frac{1}{2}} \sqrt{\gamma} A_t \\
 &= \frac{p_0 A_t \sqrt{\gamma}}{\sqrt{RT_0}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} = 71.20 \text{ kg/s}
 \end{aligned}$$

$$\text{(b)} \quad \frac{A_{\text{exit}}}{A_t} = 2 \quad \therefore M_{\text{exit}} = 2.20$$

$$\text{(c)} \quad \frac{p_{\text{exit}}}{p_{01}} = \frac{1}{10.693} \quad \therefore p_{\text{exit}} = 28056 \text{ Pa}$$

$$\text{(d)} \quad \text{For } \frac{A_{\text{exit}}}{A_t} = 2, \quad 1.0644 < \frac{p_{01}}{p_{\text{exit}}} < 10.693$$

$$\therefore \frac{1}{10.693} < \frac{p_{\text{exit}}}{p_{01}} < \frac{1}{1.0644}$$

$$\therefore 28055 \text{ Pa} < p_{\text{exit}} < 281849 \text{ Pa}$$

3.5 Wind tunnel design

★★★ 25-30 mins

A blow-down wind tunnel is designed for a Mach number of $M = 3$. The reservoir pressure is 10 bar.

- (a) What will be the pressure in the test section? (Give 3 decimal places)
- (b) What will be the pressure at entry to the diffuser, assuming a normal shock wave at the end of the test section? (Give 3 decimal places)
- (c) A fixed area second throat is to be designed for the wind tunnel. What will be its minimum cross-sectional area, determined from flow starting considerations, if the test section has a cross section 8 cm x 4 cm? (Give 2 significant figures)
- (d) What will be the Mach number at the second throat once the tunnel is running steadily? (Give 2 decimal places)

$$(a) \frac{p_1}{p_0} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}} = 0.02722$$

$$\therefore p_1 = 0.272 \text{ bar} = 27223.604 \text{ Pa}$$

(b) Normal shock wave at the end of the test section

$$\frac{p_2}{p_1} = \frac{2\gamma M^2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} = \frac{31}{3}$$

$$\therefore p_2 = 2.813 \text{ bar} = 281311.401 \text{ Pa}$$

$$(c) A_1 = 8 \text{ cm} \times 4 \text{ cm} = 32 \text{ cm}^2 = 3.2 \times 10^{-3} \text{ m}^2$$

$$\therefore M_2^2 = \frac{2 + (\gamma-1) M^2}{2\gamma M^2 - (\gamma-1)} = \frac{7}{31} \quad \therefore M_2 = \sqrt{\frac{7}{31}}$$

$$\therefore \left(\frac{A_1}{A^*}\right)^2 = \frac{1}{M_2^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_2^2\right) \right]^{\frac{\gamma+1}{\gamma-1}} = 1.9332$$

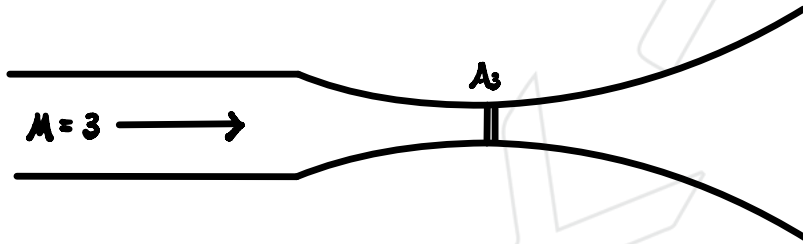
$$\therefore A^* = \frac{A_1}{\sqrt{1.9332}} = 2.3015 \times 10^{-3} \text{ m}^2$$

Let $M_0 \leq 0.99$,

$$\left(\frac{A_2}{A^*}\right)^2 = \frac{1}{M_0^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_0^2 \right) \right]^{\frac{\gamma+1}{\gamma-1}} \geq 1.0002$$

$$\therefore A_2 = \sqrt{1.0002} A^* \geq 2.3017 \times 10^{-3} \text{ m}^2$$

(d) At steady state, to minimise the running power,
shock immediately after second throat



$$\therefore A_2 = 2.3017 \times 10^{-3} \text{ m}^2$$

$$\begin{aligned} \therefore \frac{A_1}{A^*} \text{ (for } M=3) &= \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{\gamma-1}} \\ &= 17.932 \end{aligned}$$

$$\therefore A^* = \frac{81}{343} A_1 = 7.557 \times 10^{-4} \text{ m}^2$$

$$\therefore \frac{A_2}{A^*} = 3.046$$

$\therefore M_3 = 2.65$ according to the tables